

Shared — CS2 Market Fundamentals & Indicator Library

Purpose: the grounding knowledge layer for *both* systems. This is the "how the CS2 market actually works" reference used to design features, calibrate thresholds, and (where useful) prime the model. System A and System B both cite this doc rather than duplicating it — same pattern as the shared signal bus: shared knowledge, separate decision logic.

Source: distilled from a Chinese-market trading crash course. Treat every rule here as a **testable hypothesis to validate in backtest net of fees**, not gospel. Most of this is discretionary human advice; §8 covers how to convert it into hard automation constraints.

Fee note: the source quotes round-trip fees of ~3–5%. On BUFF a single side is ≈2.5%; model *actual* per-venue, round-trip fees in all EV math (System B §5 / System A §5). The principle "always deduct fees before calling something profitable" is non-negotiable.

1. Glossary (CN → EN, with model meaning)

CN term	English	Meaning / how we use it
左侧 / 右侧	Left-side / Right-side	Left = buy as price falls (staged dip-buying); Right = buy as momentum rises. Left ≈ System B's native style; Right ≈ System A's.
下轨 / 中轨 / 上轨	Lower / Middle / Upper Bollinger Band	Support / trend-strength / resistance. See §3.1.
出红柱 / 出绿柱	Red / Green volume bar	Red = selling pressure; Green = buying pressure. Used as confirmation with band touches.
存世量	Circulating supply	Total units in existence. Core selection factor (§4).
有效求购	Valid buy orders	Bids <i>near</i> market price (not lowballs). Depth/demand proxy; a mandatory selection filter (§4).
CD	Cooldown	7-day trade lock (T+7). Gates every add/exit decision.
层 / 仓	Layer / Position	1 layer = 10% of allocated capital. "5层仓" = 50% deployed (§5).
千百战	Mid-tier primary weapons	~1,000–3,000 CNY skins (M4/AK/A1).
小件 / 大件	Small / Large items	Small = 50–500 CNY; Large = expensive gloves/knives/collection skins. Regime-dependent (§7).
庄 / 盘主	Market maker / Whale	Large player manipulating price.
一波流	One-wave pump-and-dump	Sharp spike then crash. Avoid entering — this is the parked pump pattern.
筹码图 / 密集区	Volume Profile / Dense zone	Holder cost-basis distribution (§3.2).
套牢盘	Bagholders / Trapped holders	Bought high; will sell into any bounce → overhead resistance.
正T / 负T / 做T	Swing (T) trading in a position	+T = buy low then sell high; –T = sell high from holdings then rebuy lower.
接盘	Being exit liquidity	Buying while whales sell to you. The thing to never be.
割肉 / 止损	Cut loss / Stop loss	Sell at a loss to preserve capital (§6).
梭哈	All-in	100% into one item. Forbidden (§5).

2. Market regime detection (gates everything)

Before any item-level decision, classify the broad market. Regime sets the position-sizing ceiling (§5) and biases both systems.

- **Bull:** most primary weapons rise steadily daily; pullbacks small and quickly recovered; buy orders plentiful, listings few, orders fill fast; even obscure items follow.
- **Bear:** primaries drift down; bounces short-lived and fade next day; listings pile up, bids thin and keep stepping lower; "even good items can't escape" — entering is catching a falling knife.
- **Sideways/range:** prices oscillate in a band; best regime for steady range-trading.

Quantify it from breadth (share of tracked primaries above their middle band / N-day MA), listing-vs-buy-order balance market-wide, and aggregate fill speed. Emit a `market_regime` ∈ {bull, bear, sideways, weak} signal both systems read.

Per-regime action (core table):

Regime	Max deployment	Action
Bull	70-80%	Hold strong items, scale up gradually.
Sideways	~50%	Trade support/resistance; steady 10-15% profits.
Bear	≤30%	Avoid buying; cut losses immediately; never average down.
Weak / low-liquidity	small items only	Spread across 5-10 low-value items; avoid large items (\$7).

3. Indicator library (the feature set)

3.1 Bollinger Bands

Three bands: Upper (resistance), Middle (trend/strength), Lower (support).

- **Middle band = the waist.** Price above → strong, holdable, first touch often bounces. Price below → weak; do **not** bottom-fish; usually slow downtrend.
- **Upper band = resistance / sell.** Touch + **green** volume bar → standard sell (reduce/take profit; capital is exiting, follow it out). Sticks to upper band with **no** green bar → strong item riding the band, **hold**.
- **Lower band = support / buy.** Touch + **red** bar + price *stops falling* → buy signal (place low resting orders). Touch with no red bar and still falling → **no signal**, can keep dropping. Rule: don't buy until it has fully plunged *and* stabilised.
- **Band width = trend speed.** Widening → trend accelerating (don't fight it — hold or exit decisively). Narrowing → entering oscillation; range-trade.

As features: `pct_b` (position within bands), `bandwidth`, `middle_band_side`, and touch-events joined with the volume-bar sign.

3.2 Volume Profile / cost distribution (筹码图)

Distribution of holders' cost basis. Tall bars = cost-concentration (dense) zone.

- **Support/resistance:** price *above* the dense zone → that zone is **support** (holders defend cost). Price *below* → dense zone is **resistance** (bagholders sell into bounces).
- **Dense-zone drift (the key read):** drifting **up** → main force accumulating/pushing → follow with small size. Drifting **down** → distribution/selling → avoid buying. No direction → chaotic, stay away.
- **Four shapes:**
 - *Low single peak* → accumulation complete; potential uptrend; hold if price stays above the peak.
 - *High single peak* → profit-taking done, retail bought the top; **high risk**, break below peak = run.
 - *Double peak* → range; trade between; volume break above upper peak opens new upside.
 - *Scattered peaks* → illiquid/crashing; forbidden zone.
- **Never use alone** — combine with supply, case price, volume-price. It shows trend, not the exact bottom.

3.3 Volume-Price relationships (5 patterns)

Critical distinction the model must encode: *Volume* (成交额, *executed trades*) ≠ *Listings* (在售量, *open sell orders*). Volume up = more buyers (bullish); listings up = more sellers (bearish).

1. Listings ↑ + price ↓ + high volume → main force **exiting** (or rare costly shakeout). Bearish.
2. Price ↓ + low volume → **shakeout** to scare weak hands, then re-accumulate. Often a setup, not a real breakdown.
3. Volume ↑ + price ↑ together → **healthiest**; real accumulation, steady push. Confirmation to follow.
4. Price ↑ + volume ↓ → **weak rally, sell**; classic one-wave pump — never enter the pump.
5. Sideways/flat → conditional entry only with conviction/information; else wait for the first move to confirm.

Compact rule from the selection section: **listings ↓ + volume ↑ = good; listings ↑ + volume ↓ = bad.**

3.4 Whale-accumulation signals (entry timing)

Three signals that a market maker is accumulating *before* a pump — buy *ahead*:

1. **Sideways + shrinking volume (横盘缩量)**: can't drop further, sellers exhausted, listings/float slowly decreasing → light float, low resistance to a pump.
2. **Volume up, price flat (放量不涨)**: executed volume spikes several× baseline but price barely moves → whale absorbing all sell orders while pacing to avoid tipping the price.
3. **Against-the-trend resilience (逆势抗跌)**: refuses to fall on bad news / broad-market drop → strong hidden bid absorbing supply; whoever holds most inventory defends the line.

Multiple signals firing at once → high-priority watchlist. Not a guarantee; a win-rate improver.

4. Item selection criteria

4.1 The five core factors

1. **Production/source (产出)**: active drop, retired, or discontinued-collection. Retired/capped supply is bullish; a likely Armory/Arsenal re-release is bearish (supply risk).
2. **Circulating supply (存世量)**: determines the price *ceiling*; capital injection determines how high it actually goes. Not "lower is always better" — too low kills liquidity.
3. **Liquidity / daily volume (流通量)**: $\geq \sim 10$ genuine trades/day is the minimum to be "liquid"; 20–40/day is healthy.
4. **Primary weapon? (主战与否)**: AK-47, M4A1-S, M4A4, USP-S, Glock, AWP = primaries. Secondary (e.g. Galil, force-buy guns) rank lower — but traditional primaries in the mid-tier band are *saturated* after this year's rally, so secondary-primaries are the fresh-edge tier.
5. **Aesthetics (颜值)**: rank within weapon category; classic art styles (Howl / Poseidon / Boom lineage) carry a premium.

4.2 Supply tiers (reconciling the two ranges in the source)

- **High-conviction "institutional-grade" pick (esp. Factory New)**: **~2,000–10,000** existing units is the sweet spot capital prefers (Hellhound $\approx 7,000$).
- **Broader safer-liquidity selection / quick-start**: **10,000–30,000** circulation.
- **Hard exclude**: supply $> 50,000$ (upside capped, too heavy to move).

4.3 Mandatory conditions (all three required)

1. Supply in target band (§4.2).
2. **Case/source price ≥ 80 CNY** (expensive opening cost → supply won't inflate easily).
3. **≥ 3 valid buy orders** near market (e.g. item at 1000 → bids at 950–980 with 4+ qty). Valid = close to market; invalid = far-below lowballs. To buy, place *between* the valid and invalid zones.

4.4 Priority ordering

- **Category priority**: 1st-gen high-end gloves $>$ 1st-gen glove materials $>$ knife "three gods" materials $\geq$ mid-tier primaries (1,000–3,000) and 2nd/3rd-gen gloves.
- **Weapon priority**: M4A4 $>$ AK-47 $\approx$ M4A1-S $\approx$ Glock $>$ others (M4A4 = highest liquidity, easiest to recover).
- **Upper/lower tier link**: if the top-tier Gold item has an active market maker, its lower-tier Red skins usually follow — but don't blindly sweep a whole tier; watch for manipulation.

5. Position sizing & money management (layers)

Golden rule: stickers/pins never full position; in a range keep ~ 5 layers (50%); chasing momentum never exceed 2 layers (20%); never casually push to high deployment.

- Divide capital into **10 layers** (1 layer = 10%).
- **Baseline**: **50% deployed**, 50% dry powder for averaging down / sudden dips.
- **Regime scaling**: bear 30% (short swings only); strong bull 70–80%.
- **Per-category cap**: $\leq 60\%$ (**6 layers**) of that category's budget in one sub-category.
- **Single item**: ≤ 6 layers (**60%**) of its allocation; momentum-chase entries ≤ 2 layers.
- **Never all-in (梭哈); never borrow; idle funds only.**
- **Core logic**: this market rewards *longevity*, *not one big win* — a single Valve update can wipe out the reckless.

Example allocation (100k CNY): mid-tier collection 30k · mid-tier primaries 20k · materials 10k · gloves 20k · emergency-dip reserve 20k.

6. Entry / add / take-profit / stop-loss

6.1 Left-side vs right-side

- **Left-side (dip-buying, staged):** exact bottom unknown → enter in stages, keep dry powder, **wait a full CD before adding**. Rises → break even; drops → add lower. Needs good info + selection understanding. (*System B default.*)
- **Right-side (momentum):** buy only on a confirmed breakout/uptrend, when the main force's signal is obvious and the trade completes within one CD. Easier but riskier. (*System A default.*)

6.2 Staged build example (5,000 CNY, one item)

Buy 2 @600 → 2 @560 → 2 @540 → wait CD; if still ~540, 2 more (≈70–80% deployed) → sell 540-batch @550, 560-batch @570. Add **only at support** (≈ –10% from first buy) and **only after the prior batch's CD clears**; never exceed 6 layers.

6.3 Take-profit / stop-loss (hard rules)

- **Take profit at +10–15%**; sell partial at resistance. Consistency beats greed.
 - **Stop loss –10% → cut immediately** (no hoping). **–15% to –20% → unconditional liquidation.**
 - **Never average down in a bear market** — the bottom is unknowable.
 - **+T / –T** swing-trading allowed within a held position for extra yield.
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7. Small vs large items & rotation (regime-dependent)

- **Weak/low-liquidity market:** large items (gloves, high knives) are a minefield — need huge capital to move, full of bagholders, every bounce gets sold. **Small items (50–500 CNY) are the safe haven:** low listings (≤150), low entry, spread across **5–10** items, place low resting orders and wait. In a weak market, *not losing is winning*.
 - **Rotation logic:** after small items run up, capital rotates to **mid-tier (1,000–3,000)** items with real fundamentals and depth. Don't chase late-stage small-item hype — pre-place low resting orders on mid-tier candidates (low listings, not yet hyped) and wait for the rotation.
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8. Translating discretionary advice → automation constraints

The source is written for a human ("check once a day, close the app, don't obsess"). For an autonomous agent, those become **hard-coded rules**:

- "Wait for CD before adding/selling" → the scheduler **cannot** act on a lot before `unlock_time`; adds to an item are blocked until the prior batch clears CD.
 - "Don't obsess / check once a day" → System B runs on a **scheduled decision cadence** (e.g. daily), not a hot loop.
 - "Don't revenge-trade after a loss" → a **cooldown/rate-limit** on re-entering an item after a stop-loss; a tripped daily-loss limit halts new buys.
 - "Never all-in / never full position" → **position-size caps enforced in the risk gate**, not left to judgment.
 - "Take profit 10–15%, stop 10%" → **bracket orders / automated TP+SL** attached at entry.
 - "Don't follow screenshots/tips" → social is **untrusted input** feeding scoring only, never a direct trigger (already in both systems).
 - "1–2 trades/month, don't overtrade" → a turnover cap per item; longevity over churn.
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9. Trade journaling → automated logging & attribution

The human "trading journal" becomes the system's **structured trade log + performance attribution**. Record per lot: entry/exit dates, item + capital + layer %, buy/sell price *including fees*, net P&L, the 3 selection reasons (which factors/signals fired), and outcome vs. thesis. Feed this back to: - re-fit factor weights (walk-forward), - measure which signals actually carry edge net of fees, - flag recurring mistakes as new risk rules.

"10 tracked trades > 30 untracked." No log = wasted experience.

10. Which system uses what

Knowledge block	System A (reactive)	System B (positional)
Regime detection (§2)	gate: pause reactive buys in bear unless durable	primary: sets deployment ceiling
Bollinger (§3.1)	band-width = trend acceleration confirmation	core entry/exit timing
Volume Profile (§3.2)	secondary	core factor/timing feature
Volume-Price (§3.3)	pattern 3 = confirm reactive entry; pattern 4 = avoid	core
Whale signals (§3.4)	secondary	core entry timing
Selection criteria (§4)	liquidity/exit-ability check on candidates	core factor model
Position sizing (§5)	momentum-chase ≤ 2 layers; caps in risk gate	full framework
Left/right-side (§6.1)	right-side default	left-side default
TP/SL (§6.3)	yes (hard brackets)	yes (hard brackets)
Small/large & rotation (§7)	minor	regime-dependent strategy
Journaling (§9)	yes	yes

11. Empirical grounding (research corpus) — *secondary to the notes above*

Precedence: §§1–10 (derived from our crash-course notes, BUFF practitioner knowledge) are **primary and govern live trading**. This section is **secondary** — the two academic studies (/research/RESEARCH_INDEX.md + local PDFs) are used to *corroborate* the notes and to hold a *backlog of additions to test later*. Where a finding just confirms a note, the note's framing wins. Findings tagged **[new]** aren't in the notes and are retained to validate on our own BUFF data before we trust them — they don't override §§1–10 today.

- **[confirms §4/System A] Update-driven structural breaks & model returns not prices.** Prices are non-stationary with breaks on balance updates; returns are stationary. Backs the notes' "updates move the market" thesis and the returns-based modeling. (Nikolaenko)
- **[confirms §3.1] Momentum + mean reversion.** The dominant predictor is deviation-from-7-day-MA (short-term momentum) with 7-day reversion behind it — i.e. exactly the notes' Bollinger `pct_b` logic. Use the **notes'** Bollinger framing; this is validation, not a replacement. Caveat: daily *direction* ceiling $\approx 0.5 R^2$ — treat as an expectations check on the notes' more optimistic signal claims. (Pettersson)
- **[new] Model class:** gradient-boosted trees / Random Forest beat linear and *beat LSTM* (weak temporal memory → deep nets underperform). Notes are discretionary and silent on ML, so adopt this with nothing to override. (Pettersson)
- **[refines § events] Two event classes.** Balance/meta updates cause real price breaks (defer to the **notes'** System-A update logic); calendar/esports events (Steam sales, Majors, tournaments, operations) move *volume, not daily price* — the paper's added distinction, used for liquidity timing only. (Nikolaenko + Pettersson)
- **[confirms §5] Fat tails (kurtosis >100).** The statistical "why" behind the notes' "survive, don't win big." Keep the **notes'** layers framework; this just justifies it. (Nikolaenko)
- **[new] Steam sale seasonality.** Semi-annual liquidation dips in Steam sale windows. Notes (BUFF/China-centric) don't flag this — keep as secondary and verify it shows on BUFF before acting. (Nikolaenko)
- **[new] Volatility is forecastable** via GARCH even when direction isn't → enables the sizing rule below. (Nikolaenko)
- **Our edge is partly better data:** both papers were limited to 2 skins or Steam median-only; our BUFF volume + buy-order + listing-depth + float feed (cs2.sh) is richer than what either had.

Volatility-targeted sizing [new — secondary]: estimate conditional volatility per item (GARCH-style); scale position size *inversely* to forecast vol and place stops/targets in vol units. A paper-derived addition — layer it on top of (not instead of) the notes' layers framework (§5).

12. Model iteration, observability & governance (how to change the model fast)

A fully-automated system *will* make flawed decisions; the goal is to make them **diagnosable and correctable in minutes, not redeploys**.

Layered design so a bad decision is traceable. Every trade decision passes through: (1) data → (2) features/signals → (3) model/score → (4) decision & risk rules → (5) execution. When something goes wrong, triage *which layer*: bad data (feed stale/divergent), bad feature (signal miscomputed), bad model (weights/drift), or bad rule (a threshold/cap). Fix at the

lowest responsible layer.

Config-driven knobs (the fast path). Factor weights, signal thresholds, position caps, TP/SL levels, regime ceilings, and the blocklist/allowlist all live in **versioned config (YAML), not code**. Adjusting behavior = editing config + reload, no logic redeploy. This is the answer to "the model is doing X, make it stop": tighten the rule or reweight the factor in config.

Decision provenance logging (the debugging substrate). Every order logs its inputs, which signals fired, the score, the rule that triggered it, and the regime — so any bad trade is *replayable* ("why did it buy that?"). This is the trade journal (§9) elevated to an observability tool.

Override & rollback. The risk gate can veto/shrink model output live; blocklist is hot-editable; models + configs are **versioned so you can revert to last-good instantly**; a single kill switch halts all buying.

Safe change promotion. Test any model/config change in **shadow / paper mode alongside live (champion-challenger)** before promoting. Never hot-swap an untested model into live capital.

Retraining & drift. Schedule walk-forward re-fits; trigger an off-cycle re-fit on a **structural-break/drift alarm** (Bai-Perron/CUSUM — the same break detection that flags balance updates) or a factor-decay alarm from the journal. Factor weights decay (e.g. the noted mid-tier-meta saturation) — re-fit, don't trust stale weights.

What's live-editable vs. needs a redeploy: *live* = config knobs, blocklist/allowlist, kill switch, regime ceilings. *Redeploy* = retraining the model, adding features, changing execution logic.

12a. Flawed-decision runbook (the fast-adjust procedure)

When the system makes a bad trade, follow this loop — most fixes are a config edit, not a redeploy:

- 1. **Pull the decision's provenance log** (§12 logging) — inputs, signals that fired, score, rule, regime. This tells you *why* it acted.
- 2. **Localize to a layer** using the symptom table below.
- 3. **Apply the lowest-layer fix**, test in shadow, then promote. Use the kill switch first if capital is actively at risk.

Symptom	Most likely layer	Fast fix (live)	If it recurs (redeploy)
Bought an obvious pumped/parabolic item	decision rule	tighten pump-detector threshold; add item to blocklist	add/retrain a manipulation classifier feature
Over-trading / churning one item	decision rule	lower turnover cap; raise re-entry cooldown	revisit signal that keeps re-firing
Sizing too large / too concentrated	risk config	lower per-item/per-category caps; tighten vol-target	—
Won't sell a loser / ignores stop	risk config	verify TP/SL brackets are attached & firing	fix bracket-attachment logic
Good signal, bad entries in a falling market	regime gate	tighten bear-regime suppression / deployment ceiling	improve regime classifier
Same feature dominates every bad call	feature/model	down-weight that factor in config	re-fit weights (walk-forward); feature review
Model good in backtest, bad live	data or overfit	check feed freshness/divergence; pause on stale data	regularize / re-fit; check look-ahead leakage
Systematically wrong since a date	drift / structural break	—	trigger off-cycle re-fit (break alarm, §12)

Rule of thumb: if the fix is "change a number" (a weight, threshold, cap, cooldown, or blocklist entry) it's a live config edit and takes minutes. If the fix is "change what the model *learns from*" (features, training data, model class) it's a retrain/redeploy. Design so the common cases are the former.

Tooling notes

- **Claude Code** is the right tool to *build and iterate* the whole stack — data pipeline, feature engineering (Pettersson's feature table is a ready starter spec), the RF/XGBoost ranker, the GARCH vol module, Bai-Perron break detection, the backtester, and the paper-trading harness. It's an agentic coding tool (terminal / IDE / desktop). Key distinction: **Claude Code is the builder/maintainer, not the deployed model** — it produces the trained-model artifact + inference code, which then runs on your own scheduler/service 24/7. In the iteration loop, a dev describes the flaw → Claude Code inspects the provenance logs, adjusts config or retrains, runs the backtest/shadow, and proposes the change. (Docs: <https://docs.claude.com/en/docs/claude-code/overview>)
- **Obsidian** is a human knowledge-management layer, not a modeling or runtime tool. Since an Obsidian vault is just a folder of markdown files, point it at `/docs` + `/research` and the team browses/links the exact same files Claude Code reads. Complementary, not competing. Not needed for the model itself.
- **"Other agents like Hermes":** you don't need a separate coding agent to build the model — Claude Code covers

building/iterating. If "Hermes" means a self-hostable open LLM (e.g. a Nous Hermes-class model), the sensible place for it is the **social-monitor's classification layer** (cheap, high-volume Chinese-text tagging), not the price model — that's a cost/hosting choice for one component, not a replacement for Claude Code. (Flagging ambiguity: confirm which "Hermes" is meant before committing.)